

Research Methods — Module 01: Systems — Study Questions

1. What is Dynamic Causal Modelling (DCM)? How does it implement Active Inference at the level of brain imaging analysis?
2. How does DCM differ from Granger causality and structural equation modeling? What does DCM's generative model provide that correlation-based methods lack?
3. What is Bayesian Model Selection (BMS)? How does it compare models of brain connectivity using free energy as the model evidence approximation?
4. Design a DCM study to investigate the neural basis of prediction error processing in the auditory cortex. What would your generative model architecture look like? What brain regions would you include?
5. What is the Parametric Empirical Bayes (PEB) framework? How does it extend DCM to group-level inference?
6. How do you choose the level of model complexity in DCM? What is the role of the free energy approximation to model evidence in preventing overfitting?
7. What is computational phenotyping? How might individual differences in Active Inference parameters (precision, prior strength, learning rate) serve as biomarkers for psychiatric conditions?
8. Design a behavioral experiment that could distinguish high-precision-prior subjects from low-precision-prior subjects. What task would you use, and what behavioral signatures would you expect?
9. What is the role of EEG/MEG in testing oscillatory predictions of Active Inference (gamma = prediction error, beta = predictions)? What temporal resolution is needed?
10. How can Active Inference models be validated using eye-tracking data? What predictions about saccadic patterns follow from expected free energy minimization?
11. What ethical considerations arise in computational psychiatry? If you could identify someone's "inference parameters" that predispose them to depression or anxiety, what are the risks of this knowledge?
12. How reproducible are DCM results? What are the main threats to reproducibility in computational modelling of brain imaging data?

13. Design a simulation study that tests whether a given Active Inference agent model can reproduce a known behavioral phenomenon (e.g., the Stroop effect, Wisconsin Card Sort Test).
14. What is model recovery analysis? Why is it essential for validating computational models — showing that your fitting procedure can distinguish Model A from Model B?
15. How might machine learning (deep neural networks trained on brain imaging data) complement DCM for Active Inference research? What are the trade-offs between data-driven and theory-driven approaches?
16. How do you validate a computational psychiatry model clinically? What evidence would demonstrate that Active Inference-based treatment recommendations improve patient outcomes?
17. What is the relationship between Active Inference and reinforcement learning models (e.g., Q-learning, actor-critic)? Can RL experiments be reanalyzed through an Active Inference lens?
18. How can you use pharmacological interventions (e.g., ketamine, psilocybin, scopolamine) to test Active Inference predictions about precision and prediction errors?
19. Design a longitudinal study tracking how Active Inference parameters change with treatment (e.g., antidepressant therapy). What parameters would you expect to change, and in what direction?
20. Critically evaluate: Is computational modelling of psychiatric conditions genuinely predictive (enabling personalized treatment), or merely descriptive (providing post-hoc mathematical accounts of symptoms)?